

MOGS-Congenital Disorder of Glycosylation (MOGS-CDG / CDG-IIb): Comprehensive Disease Characteristics Report

MONDO: MONDO:0011629 | **Category:** Mendelian (autosomal recessive inborn error of metabolism) **Prepared:** Iterations 1–5 — evidence base combines primary literature (human clinical case series and mechanistic studies) with live public-database queries (gnomAD constraint, ClinVar variant spectrum, UniProt protein architecture, ClinicalTrials.gov). MOGS-CDG is ultra-rare (~30 patients reported worldwide as of 2024), so most clinical evidence is from individual patients and small case series rather than aggregated registry-scale data.

Evidence provenance (database queries performed): gnomAD (constraint: pLI $\approx$ 0, LOEUF = 0.84 $\rightarrow$ LoF-tolerant, recessive) · ClinVar (647 *MOGS* records; $\approx$ 27 P/LP vs 153 VUS; overwhelmingly SNVs) · UniProt Q13724 (837-aa type II ER membrane GH63 enzyme; catalytic D583/E807; N-glycosylated at N657) · ClinicalTrials.gov (no MOGS-CDG-specific interventional trials). Evidence types are labelled throughout as human-clinical, in vitro, model-organism, or computational/database.

1. Disease Information

Overview. MOGS-CDG is an ultra-rare autosomal recessive congenital disorder of glycosylation caused by biallelic loss-of-function variants in *MOGS*, which encodes **mannosyl-oligosaccharide glucosidase (glucosidase I / GCS1)**, the first enzyme in the endoplasmic-reticulum (ER) processing/trimming of N-linked oligosaccharides. Loss of glucosidase I activity blocks the initial trimming of the Glc₃Man₉GlcNAc₂ N-glycan precursor, disrupting N-glycan maturation on many glycoproteins. It is classified as a **CDG type II** (a defect of glycan *processing/remodeling*, as opposed to type I assembly defects). The first patient was described by De Praeter et al. in 2000

([P 10788335](#)).

Key identifiers: - **OMIM:** #606056 (Congenital disorder of glycosylation, type IIb) — gene *MOGS* OMIM 601336 - **Orphanet:** ORPHA:79328 (MOGS-CDG / CDG-IIb) - **Mondo:** MONDO:0011629 - **ICD-10:** E77.8 (other disorders of glycoprotein metabolism); **ICD-11:** 5C51.2 (disorders of N-glycosylation) - **MeSH:** Congenital Disorders of Glycosylation (D018981) - **Gene HGNC:** *MOGS* (HGNC:24862); UniProt Q13724 (*MOGS_HUMAN*)

Synonyms / alternative names: - CDG-IIb / CDG type IIb - Glucosidase I deficiency; GCS1-CDG - Mannosyl-oligosaccharide glucosidase deficiency - Congenital disorder of glycosylation type 2b

Data source type: Individual patients / small international case series (EHR- and research-derived), not population registry aggregates, reflecting the disorder's rarity.

2. Etiology

Primary cause (genetic). Biallelic (homozygous or compound heterozygous) pathogenic variants in *MOGS* (chromosome 2p13.1) causing near-complete loss of glucosidase I enzymatic activity. This is a monogenic Mendelian defect; there are no established environmental or infectious causes.

- De Praeter (2000,

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first patient was a compound heterozygote for missense variants **R486T** and **F652L**; residual enzyme activity <3% of controls; both parents ~50% (obligate carriers) → autosomal recessive.

- Völker (2002,

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glucosidase I activity <1% of control in patient fibroblasts with intermediate parental values.

Genetic risk factors. The only risk factor is inheritance of two pathogenic *MOGS* alleles. **Consanguinity** raises risk of homozygous forms; reported cases are from diverse populations (European, Chinese, Japanese, Korean, Indian), consistent with private/family-specific variants rather than a common founder allele.

Environmental risk factors. None identified — disease is fully determined by genotype (congenital, present at birth).

Protective factors. No genetic or environmental protective factors are established. Notably, the glycosylation defect confers an *in vitro* protective phenotype against certain N-glycosylation-dependent enveloped viruses (see Mechanism/Immune), but this is not a clinically established protective factor.

Gene–environment interactions. Not applicable in a classical sense; disease expression is genotype-driven. Phenotypic variability among patients with similar genotypes suggests modifier effects (genetic background, residual enzyme activity), but specific modifiers are unidentified.

3. Phenotypes

MOGS-CDG is a multisystem disorder with prominent **neurological** involvement. Frequencies below are qualitative given the small cohort (~30 patients; Teutonico 2024,

P 38498292	Post	2023,
P 36651519	Shimada	2022,
P 35790351		).

Phenotype	Type	Onset	Frequency	HPO term
Muscular hypotonia	Clinical sign	Neonatal/congenital	Nearly universal	HP:0001252
Global developmental delay / intellectual disability	Clinical sign	Infancy	Nearly universal	HP:0001263 / HP:0001249
Seizures / epileptic encephalopathy (often drug-resistant)	Clinical sign	Early infancy	Nearly universal	HP:0001250 / HP:0200134
Feeding difficulties / failure to thrive	Symptom	Neonatal	Common	HP:0011968 / HP:0001508
Hepatomegaly / hepatic dysfunction (elevated transaminases)	Sign/lab	Neonatal–infancy	Common	HP:0002240 / HP:0001392
Dysmorphic facies (long eyelashes, retrognathia, hirsutism, depressed nasal bridge, high palate, blepharophimosis)	Physical	Congenital	Common	HP:0000527, HP:0000278, HP:0001007
Clenched/overlapping fingers, overlapped toes	Physical	Congenital	Common	HP:0001188
Hypoventilation / respiratory insufficiency	Sign	Neonatal	Common (severe cases)	HP:0002791
Generalized edema / abnormal fat distribution	Sign	Neonatal	Reported subset	HP:0007430
Movement disorder (dystonia, hyperkinetic movements)	Sign	Infancy–childhood	Subset (older survivors)	HP:0001332 / HP:0002072
Hypogammaglobulinemia / immunodeficiency	Lab	Infancy	Subset	HP:0002720
Progressive cerebral/cortical–subcortical atrophy, thin corpus callosum, ventricular dilation	Imaging	Infancy	Common	HP:0002059 / HP:0007371

Phenotype	Type	Onset	Frequency	HPO term
Vision problems	Symptom	Variable	Minority	HP:0000505
Nephromegaly, hypothyroidism, GERD, auditory neuropathy, Hirschsprung disease	Sign	Variable	Rare/ individual reports	HP:0000105, HP:0000821, HP:0002020, HP:0002232

Severity/progression. Variable, ranging from **fatal in infancy** (first patient died at 74 days,

[P 10788335](#) to **survival into adulthood** (oldest reported 19 years,

[P 33058492](#)).

Neurological course is generally **progressive** (worsening encephalopathy, brain atrophy) with drug-resistant epilepsy.

Quality-of-life impact. Profound: severe neurodevelopmental disability, drug-resistant seizures, feeding/respiratory support needs, and dependence for daily activities. No validated disease-specific QoL instruments exist for this ultra-rare disorder.

4. Genetic / Molecular Information

- **Causal gene:** *MOGS* (mannosyl-oligosaccharide glucosidase; a.k.a. glucosidase I, GCS1). HGNC:24862; NCBI Gene 7841; OMIM 601336; UniProt Q13724; **Ensembl** ENSG00000115275. Reference transcript NM_006302. Located 2p13.1; a compact gene (**chr2:74,461,057–74,465,410, GRCh38**, ~4.35 kb) encoding an **837-aa, ~91.9 kDa type II single-pass ER membrane glycoprotein** of glycoside hydrolase family **CAZy GH63** (EC 3.2.1.106).
- **Population constraint (gnomAD, GRCh38):** *MOGS* is **loss-of-function tolerant at the heterozygous level** — **pLI** ≈ 0 (3.9×10^{-10}), observed/expected LoF **oe_lof** = **0.66** (**LOEUF** = **0.84**), missense **Z** = 0.97 (**oe_mis** = 0.93). This means haploinsufficiency is not disease-causing and healthy carriers are expected — exactly the signature of a **recessive enzymopathy** requiring biallelic loss of function (matches the ~50% carrier enzyme activity in healthy parents;

[P 12145188](#)

- **Pathogenic variant spectrum** — predominantly **missense**, with some frameshift/duplication:

- R486T (c.1587G>C) and F652L — original compound heterozygote

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- c.1239_1267dup (p.Asp414Leufs*17), c.544G>A (p.Gly182Arg), c.1698C>A (p.Asp566Glu) in Chinese siblings

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- Numerous additional private missense/compound-heterozygous variants across ~30 patients (Shimada 2022

P 35790351;

Teutonico 2024

P 38498292)

- **Variant classification (ACMG/AMP):** Pathogenic/Likely pathogenic when biallelic with functional confirmation (enzyme assay or urine Glc₃Man). Structural modeling has been used to support pathogenicity

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ClinVar landscape (Iteration 3 query): 647 MOGS records; of ~300 with classifications — **16 Pathogenic, 9 Likely pathogenic, 2 P/LP (≈27 P/LP), 1 Conflicting, 153 VUS, 114 Likely benign.** The large VUS fraction highlights the diagnostic importance of orthogonal functional confirmation (urine Glc₃Man, enzyme assay, or yeast *CWH41* complementation) to resolve novel variants.

- **Variant type distribution (ClinVar):** Overwhelmingly **single-nucleotide variants (278)** with a minority of small deletions (11), duplications (8), copy-number losses (2), and insertions (1) — i.e., predominantly missense and small frameshift alleles; structural variants are rare (so CMA/karyotype/FISH have little diagnostic role).
- **Variant types/class:** missense (most common), frameshift/duplication, and predicted splice/nonsense in some cases → **loss of function.**
- **Allele frequency:** Individually extremely rare/absent in gnomAD (private variants); no common recurrent allele; carrier frequency very low.

- **Somatic vs germline:** Germline only.
- **Functional consequence:** Loss of function (near-complete abolition of glucosidase I catalytic activity, <1–3% residual).
- **Modifier genes:** None confirmed. Compensatory upregulation of **endo- α 1,2-mannosidase (MANEA)** partially bypasses the block (P 12145188) and may modulate phenotype.
- **Epigenetic information:** None reported.
- **Chromosomal abnormalities:** None; point/small variants only.

5. Environmental Information

- **Environmental factors:** None causally implicated. Disease is congenital and genotype-determined.
- **Lifestyle factors:** Not applicable (disease presents at/near birth).
- **Infectious agents:** Not a cause. Of note, patients show *reduced* susceptibility to certain enveloped viruses in vitro because viral entry glycoproteins depend on host N-glycosylation (P 24716661).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic loss-of-function variants in *MOGS*** → results in near-complete deficiency (<1–3% residual) of ER glucosidase I. (*Demonstrated: enzyme assays,*

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2. **Loss of glucosidase I activity** → fails to cleave the distal $\alpha 1,2$ -linked glucose from the protein-bound **Glc₃Man₉GlcNAc₂** N-glycan precursor, the first committed step of N-glycan trimming in the ER. (*Demonstrated.*)
3. **Blocked trimming** → leads to accumulation of non-deglucosylated high-mannose N-glycans (**Glc₃Man₇₋₉GlcNAc₂**) on nascent glycoproteins and diversion of the precursor through **endo- $\alpha 1,2$ -mannosidase**, releasing the free tetrasaccharide **Glc₃Man**, which is excreted in urine (diagnostic biomarker). (*Demonstrated,*

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4. **Impaired N-glycan maturation** → disrupts the **calnexin/calreticulin glycoprotein quality-control cycle** (which depends on monoglucosylated glycans) and downstream complex/sialylated glycan formation on many secreted and membrane glycoproteins. (*Partly inferred from glycan-processing biology; abnormal serum/IgG N-glycomes demonstrated,*

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5. **Aberrant glycoprotein processing** → branches into multiple organ effects:
6. **Neurological branch:** abnormal glycosylation of neuronal ion channels, adhesion molecules, and receptors → **developmental & epileptic encephalopathy, hypotonia, progressive cerebral atrophy, movement disorder**. (*Inferred; clinical correlation strong,*

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7. **Immune branch:** aberrant IgG glycosylation and **shortened immunoglobulin half-life** → **hypogammaglobulinemia**; simultaneously, host-glycan-dependent enveloped viruses (HIV, influenza) show impaired entry/replication → **paradoxical relative viral resistance despite low Ig**. (*Demonstrated in vitro,*

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8. **Hepatic branch:** abnormal glycoprotein handling → **hepatomegaly, elevated transaminases, coagulopathy** (glycosylated clotting factors). (*Inferred/clinical.*)
9. **Dysmorphogenesis branch:** disrupted glycosylation during development → **craniofacial dysmorphism, digital anomalies, edema.** (*Inferred.*)
10. **Cumulative multisystem glycoprotein dysfunction** → **progressive neurodegeneration, feeding/respiratory failure, and (in severe cases) death in infancy;** milder residual function permits survival into childhood/adolescence.

Category detail: - **Molecular pathways:** N-linked glycan biosynthesis/processing (KEGG hsa00510); ER glycoprotein quality control / calnexin cycle (Reactome "Calnexin/calreticulin cycle", R-HSA-901042; "Asparagine N-linked glycosylation" R-HSA-446203). - **Cellular processes:** ER protein processing, glycoprotein folding/quality control, ER-associated degradation modulation; neuronal excitability dysregulation (epilepsy). - **Protein dysfunction:** Loss of function of glucosidase I. **Protein architecture (UniProt Q13724):** an 837-residue, ~91.9 kDa **type II single-pass ER membrane glycoside hydrolase** of **CAZy family GH63** (EC 3.2.1.106), with a short cytoplasmic tail (1–38), transmembrane signal-anchor (39–59), and a large luminal catalytic domain (60–837; residues 76–137 required for ER targeting). Catalysis uses a **proton donor at position 583 and proton acceptor at 807**; the enzyme is itself N-glycosylated at Asn657. Pathogenic missense residues (e.g., **R486, F652**;

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map to this luminal catalytic domain near the active-site machinery, disrupting acid–base catalysis and yielding <1–3% residual activity — loss of function by catalytic-domain disruption rather than aggregation. - **Metabolic changes:** Accumulation and urinary excretion of free oligosaccharide **Glc₃Man**; hypermannosylated glycopeptides; dyslipidemia and elevated CK reported

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- **Immune involvement:** Hypogammaglobulinemia (shortened IgG half-life), reduced T/NK proportions, complement C3/C4 deficiency, elevated IL-6, yet reduced enveloped-virus susceptibility

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- **Biochemical abnormality:** Enzyme deficiency of glucosidase I (EC 3.2.1.106; CAZy GH63). Catalytic reaction (UniProt Q13724): hydrolysis of the distal α-1,2-glucose from protein-bound Glc₃Man₉GlcNAc₂, releasing β-D-glucose — Reactome R-HSA-4793954 ("Glucosidase I removes

glucose from N-glycan"). - **Molecular profiling:** Serum/IgG N-glycomics show non-deglucosylated Glc₃Man₇₋₉GlcNAc₂ glycans and reduced core-fucosylated complex IgG glycans (

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glycomics show compensatory increase in Man₅GlcNAc₂ (

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Suggested ontology terms: GO:0006487 (protein N-linked glycosylation), GO:0009311/GO:0006491 (oligosaccharide/N-glycan processing), GO:0004573 (mannosyl-oligosaccharide glucosidase activity), GO:0005788 (ER lumen). Cell types: CL:0000540 (neuron), CL:0000182 (hepatocyte), CL:0000786 (plasma cell). CHEBI: CHEBI:59080-class oligosaccharides; glucose (CHEBI:17234).

7. Anatomical Structures Affected

- **Organ level (primary):** Brain/CNS (UBERON:0000955), liver (UBERON:0002107).

Secondary: immune system (UBERON:0002405), skeletal muscle/neuromuscular (hypotonia), heart (pericardial effusion –

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kidney (nephromegaly), larynx (laryngomalacia/subglottic stenosis).

- **Body systems:** Nervous, hepatic/digestive, immune, respiratory, musculoskeletal.
- **Tissue/cell level:** Neurons and neural tissue (progressive cortical/subcortical atrophy); hepatocytes; plasma cells/B lymphocytes (Ig production); broadly any secretory/membrane-glycoprotein-producing cell.
- **Subcellular level: Endoplasmic reticulum** (UBERON n/a; GO:0005783 / GO:0005788 ER lumen) — site of glucosidase I action; secretory pathway broadly affected.
- **Localization / lateralization:** Bilateral, symmetric CNS involvement (diffuse cerebral atrophy). Systemic/generalized rather than focal.

8. Temporal Development

- **Onset: Congenital / neonatal** — hypotonia, dysmorphism, feeding and respiratory problems from birth; seizures in early infancy.
- **Onset pattern:** Subacute-to-chronic with early neonatal symptoms and progressive deterioration.
- **Progression:** Generally **progressive** neurodegeneration (worsening encephalopathy, cortical/subcortical atrophy). Some manifestations evolve (epileptic spasms in infancy ceasing by ~age 7 while tonic seizures persist; emergence of dystonia/hyperkinetic movement disorder —

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- **Course/duration:** Chronic, lifelong. Severe forms are lethal in infancy (74 days in index case); milder forms survive to adolescence/adulthood.
- **Remission:** No spontaneous remission; seizures are typically drug-resistant. Certain seizure types may abate with age.
- **Critical period:** Neonatal/early-infancy window is critical for diagnosis and supportive intervention.

9. Inheritance and Population

- **Epidemiology:** Ultra-rare; ~30 patients reported worldwide as of 2024

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Prevalence/incidence not formally estimated (Orphanet: <1/1,000,000); no reliable per-100,000 figures.

- **Inheritance: Autosomal recessive** (both parents obligate carriers with ~50% enzyme activity;

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- **Penetrance:** Complete for biallelic loss-of-function genotypes.
- **Expressivity: Variable** (lethal infancy → survival to adulthood), even within families.

- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism / founder effects:** None documented; variants are private/family-specific.
- **Consanguinity:** Contributes to homozygous cases in consanguineous families.
- **Carrier frequency:** Very low (private variants; largely absent from gnomAD). Consistent with this, gnomAD shows MOGS is LoF-tolerant ($pLI \approx 0$, $LOEUF = 0.84$), so heterozygous carriers are asymptomatic and population-frequent enough to be observed, but biallelic combinations are exceedingly rare.
- **Population demographics:** Reported across European, Chinese, Japanese, Korean, and Indian families; no ethnic predilection or geographic clustering. No clear sex bias (both sexes affected). Age distribution: presents in neonates/infants.

10. Diagnostics

- **First-line clue:** Multisystem neonatal presentation (hypotonia, seizures, dysmorphism, hepatomegaly) with **normal serum transferrin isoelectric focusing** — MOGS-CDG escapes routine CDG transferrin screening

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- **Key biochemical test:** Urine oligosaccharide analysis (MALDI-TOF MS) detecting the pathognomonic free tetrasaccharide **Glc₃Man** — reliable screening/confirmation

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Quantitative Glc₃Man assays now available.

- **Enzyme assay:** Glucosidase I activity in fibroblasts/liver markedly reduced (<1–3%) — historical gold standard

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- **Glycomics:** Serum/IgG N-glycan profiling shows accumulated $\text{Glc}_3\text{Man}_{7-9}\text{GlcNAc}_2$ and altered IgG glycans

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- **Genetic testing (definitive): Whole-exome sequencing** is the principal diagnostic route in most recent cases

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WGS, targeted CDG gene panels, or single-gene *MOGS* sequencing also applicable. CMA/karyotype/FISH/mtDNA/repeat testing **not** indicated. Confirm biallelic *MOGS* variants + segregation.

- **Imaging:** Brain MRI — cerebral/cortical-subcortical atrophy, thin corpus callosum, ventricular dilation.
- **Supportive labs:** Low immunoglobulins/hypogammaglobulinemia, elevated transaminases, coagulopathy, dyslipidemia, elevated CK, complement C3/C4 deficiency

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- **Differential diagnosis:** Other CDG type II subtypes, other early-infantile epileptic/developmental encephalopathies, other inborn errors with hypotonia + hepatopathy + dysmorphism; distinguished by normal transferrin IEF + urine Glc_3Man + *MOGS* genotype. UGGT1-CDG

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is a related ER quality-control CDG in the differential.

- **Screening:** Not on standard newborn screening panels; cascade carrier testing feasible once a familial variant is known; prenatal/PGT possible for known familial variants.

11. Outcome / Prognosis

- **Survival/mortality:** Highly variable and often poor. Index patient died at **74 days**

(P 10788335);

severe neonatal forms with hypoventilation/failure to thrive carry high infant mortality.

Milder cases survive into childhood/adolescence; **oldest reported 19 years**

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No formal 5-/10-year survival statistics exist.

- **Morbidity/function:** Severe neurodevelopmental disability, drug-resistant epilepsy, hypotonia, feeding and respiratory dependence, movement disorder in survivors — profound long-term disability.
- **Complications:** Recurrent infections despite paradoxical viral resistance, aspiration/respiratory failure, hepatic dysfunction/coagulopathy, status epilepticus, failure to thrive.
- **Recovery potential:** No cure; management is supportive. Neurological damage is largely irreversible/progressive.
- **Prognostic factors:** Degree of residual enzyme activity, severity of neonatal respiratory/feeding compromise, seizure control. No validated molecular prognostic biomarkers; urinary Glc₃Man is diagnostic rather than prognostic.

12. Treatment

No disease-specific or curative therapy exists. Management is multidisciplinary and supportive.

- **Pharmacotherapy:** **Anti-seizure medications** for epilepsy (often drug-resistant; combinations frequently required) — NCIT anticonvulsant agents (NCIT:C264). **Immunoglobulin replacement (IVIG)** for symptomatic hypogammaglobulinemia (NCIT:C29099). Nutritional support, management of reflux/hepatic/coagulation issues.
- **Advanced/experimental therapeutics:** No approved gene, cell, RNA, or enzyme-replacement therapy. The α -glucosidase inhibitor **miglustat** (an iminosugar) has been studied in relation to the MOGS glycosylation phenotype and antiviral glycan modification, but **not** as a disease-modifying treatment for MOGS-CDG itself

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Miglustat NCIT:C61765. A **ClinicalTrials.gov API query (Iteration 3)** for "MOGS-CDG / glucosidase I deficiency / CDG-IIb" returned **no interventional trials specific to MOGS-CDG** (keyword hits were unrelated glucosidase disorders — Pompe/Gaucher ERT and gene-therapy studies), confirming the absence of a disease-specific experimental therapeutic pipeline to date.

- **Surgical/interventional:** Supportive only (e.g., gastrostomy for feeding, respiratory support).
- **Supportive/rehabilitative:** Physical, occupational, and speech therapy; feeding support; respiratory care; developmental/palliative care.
- **Treatment outcomes:** Symptomatic benefit only; seizures frequently refractory. No response-rate data given rarity.
- **Personalized medicine:** Care guided by organ involvement; genetic counseling for families.
- **Pharmacogenomics:** None established.

13. Prevention

- **Primary prevention:** Not possible (congenital genetic disorder). **Genetic counseling** for at-risk families; carrier screening for relatives; **prenatal diagnosis** and **preimplantation genetic testing (PGT)** available when the familial *MOGS* variants are known.
 - **Secondary prevention:** Early recognition via urine oligosaccharide analysis (Glc₃Man) and WES in neonates with encephalopathy/dysmorphism/hepatopathy, enabling timely supportive care.
 - **Tertiary prevention:** Aggressive seizure management, IVIG to reduce infections, nutritional/respiratory support to limit complications.
 - **Immunization/public health/environmental measures:** Not applicable to disease causation; standard immunizations and infection precautions apply, individualized given the immune phenotype.
 - **Counseling:** Autosomal recessive recurrence risk 25% per pregnancy for carrier couples — central genetic counseling message.
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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *MOGS* is evolutionarily conserved. Orthologs: mouse *Mogs* (NCBI Gene 57377), and the yeast homolog **CWH41/GLS1** in *Saccharomyces cerevisiae* (NCBI Taxon 4932). A prokaryotic MOGS (pMOGS) was recently identified in *Elizabethkingia meningoseptica*

([P 40674822](#)).

- **Natural disease in animals:** No well-characterized naturally occurring MOGS-CDG equivalent reported in companion animals or wildlife (OMIA has no established entry analogous to human MOGS-CDG). Veterinary relevance limited.
- **Comparative biology:** The N-glycan trimming pathway and glucosidase I function are highly conserved from yeast to humans, making cross-species functional assays informative (see Model Organisms).
- **Transmission:** Not applicable (non-infectious, non-zoonotic).

15. Model Organisms

- **Yeast (*S. cerevisiae*):** *CWH41* (MOGS homolog) knockout strains provide a **functional complementation assay** — human *MOGS* and its disease variants can be tested for rescue of the N-glycan profile, allowing pathogenicity assessment of patient mutations

([P 40674822](#)).

Strong tool for variant interpretation.

- **Prokaryotic MOGS:** A bacterial MOGS (pMOGS) characterized as an additional model/reagent for studying MOGS activity

([P 40674822](#)).

- **Cellular / in vitro models:** Patient-derived **skin fibroblasts** (enzyme kinetics, glycan analysis;

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MOGS-null transfected cells used for glycan/antiviral studies

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overexpression systems for glucosidase-inhibitor and viral-glycoprotein studies

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- **Mammalian genetic models:** No widely reported viable *Mogs* knockout mouse disease model in the retrieved literature; complete loss is expected to be developmentally severe. This is a **limitation** — no established rodent model recapitulating the human neurological phenotype.
- **Model characteristics/limitations:** Yeast/cellular systems faithfully recapitulate the **biochemical** defect (glycan trimming block) and are excellent for variant functional testing, but do **not** reproduce the multisystem neurological/immunological human phenotype. Applications: variant pathogenicity assays, glycan pathway/compensation studies, and antiviral glycan-modification research.
- **Resources:** SGD (yeast CWH41), MGI (mouse *Mogs*), Cellosaurus (patient fibroblast lines).

Key Supported vs. Refuted Hypotheses

Supported: - MOGS-CDG is autosomal recessive, caused by biallelic loss-of-function *MOGS* variants abolishing glucosidase I activity

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 - The disorder escapes transferrin-based CDG screening; urine Glc₃Man is the diagnostic biomarker

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 - Core phenotype = neonatal hypotonia + developmental/epileptic encephalopathy + dysmorphism + hepatic dysfunction, with a paradoxical immune signature

(hypogammaglobulinemia + relative viral resistance)
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Refuted / not supported: - Not detectable by standard serum transferrin IEF (normal pattern) — refutes reliance on transferrin screening. - No environmental/infectious cause; not associated with chromosomal abnormalities, somatic mutation, founder alleles, or repeat expansion.

Limitations and Future Directions

- Evidence rests on ~30 individual cases — no registry-scale epidemiology, natural-history, or QoL data; frequencies are qualitative.
- No approved disease-modifying therapy; enzyme/gene/substrate strategies unexplored clinically. Genotype–phenotype correlations and modifiers (e.g., endo- α 1,2-mannosidase compensation) merit study.
- A faithful mammalian disease model is lacking; yeast/cellular complementation assays are the current functional standard.
- The therapeutic and antiviral implications of the MOGS glycan phenotype (miglustat, host-glycan–dependent viruses) warrant continued investigation.

Primary References (PMID)

10788335 (De Praeter 2000, first case) · 12145188 (Völker 2002, enzymology/compensation) · 24716661 (Sadat 2014, immune phenotype/viral resistance) · 29235540 (Kim 2018, dysmorphism) · 30587846 (Li 2019, compound het variants) · 33058492 (Lo Barco 2021, oldest patient/movement disorder) · 33245474 (Nunes-Santos 2021, miglustat/glycan) · 33261925 (Anzai 2021, normal transferrin/urine oligosaccharides) · 35137040 (Beimdiek 2022, serum/IgG N-glycomics) · 35790351 (Shimada 2022, clinical/biochemical/genetic characterization) · 36158009 (Abuduxikuer 2022, updated clinical/glycomic) · 36651519 (Post 2023, diagnostic Glc₃Man quantitation) · 38498292 (Teutonico 2024, review, ~30 patients) · 40674822 (Zou 2025, prokaryotic MOGS/yeast complementation) · 41192964 (Shwetabh 2025, DEE case) · 40267907 (Dardas 2025, related UGGT1-CDG).

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